



POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE	SCHOOL INFRASTRUCTURE & ENROLLMENT ANALYTICS DASHBOARD
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PROJECT CATEGORY	Education Analytics School Analytics Infrastructure Analytics
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INSTITUTE NAME	Infinity Code Nexus
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TRAINER / GUIDE	Anamaina Anand Kumar
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CREATED BY	Dasari Pranitha Sri
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TOOLS USED	Power BI Desktop Power Query, DAX & Power BI Service
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SUBMISSION DATE	[13/06/2026]
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Prepared as part of the Power BI Masterclass Program to demonstrate skills in
Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX,
Storytelling, and Business Intelligence Reporting.

PROJECT OVERVIEW

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PROJECT OBJECTIVE

Develop a comprehensive Power BI dashboard to analyze school infrastructure, enrollment, teacher availability, and educational distribution across locations.

Participants are expected to:

- ✓ Analyze school infrastructure
- ✓ Evaluate student enrollment
- ✓ Assess teacher availability
- ✓ Compare school categories
- ✓ Generate meaningful insights
- ✓ Present recommendations

PROJECT GOAL

Convert raw school data into actionable insights that help understand educational facilities, student distribution, and school performance across different regions.

BUSINESS PROBLEM

Educational departments and school administrators need visibility into school infrastructure, student enrollment, and resource allocation.

The objective is to identify:

- Which regions have more schools?
- Which schools have higher enrollment?
- How are teachers distributed?
- Whether infrastructure is adequate?
- How school categories vary by region?

BUSINESS QUESTIONS

Participants must answer:

- 1. How many schools are available?
- 2. Which districts have more schools?
- 3. Which school categories dominate?
- 4. What is the total student enrollment?
- 5. What is the teacher-to-student ratio?
- 6. Which schools have better infrastructure?
- 7. Which regions need improvement?

DATASET INFORMATION

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DATASET NAME : Basic Details of Schools
DATASET DETAILS : Approximate Records 500+
Columns : 44

EXPECTED ANALYSIS AREAS

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School Analytics
Enrollment Analytics
Teacher Analytics
Infrastructure Analytics
Location Analytics

POWER QUERY TRANSFORMATIONS

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DATA CLEANING ACTIVITIES

- ✓ Removed Duplicate Records
- ✓ Validated School Data
- ✓ Handled Missing Values
- ✓ Corrected Data Types
- ✓ Standardized School Categories

DATA PREPARATION ACTIVITIES

- ✓ Renamed Columns
- ✓ Created Custom Columns
- ✓ Replaced Values
- ✓ Split Columns
- ✓ Removed Unnecessary Columns

CUSTOM COLUMNS CREATED

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Student Enrollment Group

Teacher Availability Category

School Category Classification

Infrastructure Status

Location Category

SCREENSHOT

FINAL PROJECT

FileHomeTransformAdd ColumnViewToolsHelp

Queries

Table.RenameColumns(#"changed Type2",{"Teachers", "Teachers in school"})

AR_C State

AR_C State Code

AR_C District

AR_C District Code

AR_C Sub-District

AR_C Sub-District Code

AR_C Cluster

AR_C

1	Telangana	36	Suryapet	696	Huzurnagar	5333	Zphs Lingagiri	A
2	Uttar Pradesh	9	Gorakhpur	148	Belghat	1171	Pipersandi	S
3	Andhra Pradesh	28	Kurnool	511	Halaharvi	5167	Zphs Gulyam	B
4	Maharashtra	27	Gondia	476	Deori	4385	Kakodi	V
5	Madhya Pradesh	23	Dindori	404	Amarpur	3829	Hss Amarpur (23410804816)	S
6	Chhattisgarh	22	Raipur	387	Arang	3694	Crc Kosrangi	K
7	Madhya Pradesh	23	Morena	417	Kailaras	3916	Hss Sujarma (23020605714)	K
8	Gujarat	24	Panch Mahals	454	Halol	4186	Talavadi	D
9	Odisha	21	Sundargarh	373	Bisra	3575	Madhusudanpali Up School	W
10	Maharashtra	27	Pune	490	Indapur	4516	Indapur No 1	Ir
11	West Bengal	19	Purba Medinipur	NULL	Chandipur	NULL	Narghat	C
12	Assam	18	Bongaigaon	281	Srijangram	2506	Pachania	P
13	Karnataka	29	Vijayapura	530	Muddebihal	5796	Ingalageri	H
14	Madhya Pradesh	23	Gwalior	407	Bhitarwar	3848	Hss Bhitarwar	B
15	Kerala	32	Kozhikode	561	NULL	NULL	Gips Poolakode	P
16	Maharashtra	27	Bid	NULL	Beed	NULL	V P Neknoor	K
17	Gujarat	24	Navsari	453	Vansda	4183	Lakhavadi	Li
18	Punjab	3	Mansa	37	Mansa	307	Gps Bhai Desa	B
19	West Bengal	19	Purba Medinipur	NULL	Haldia Municipality	NULL	Sutahata North	H
20	Uttar Pradesh	9	Amethi - Csm Nagar	NULL	Amethi	NULL	Kherauna	H
21	Jharkhand	20	Ranchi	339	Kanke	3223	Ms Bariyatu	W
22	Himachal Pradesh	2	Kangra	18	Nagrota Bagwan	158	Gps Malan	M
23	Uttar Pradesh	9	Agra	118	NULL	NULL	Tanora Noorpur	P
24	Manipur	14	Kangpokpi	712	Kangpokpi	2370	Maram Junior High School	N
25	Telangana	36	Suryapet	696	Ananthagiri	7226	Zphs Shanthinagar	G
26	Bihar	10	Muzaffarpur	208	Aurai	1878	M.S Ratwara	R
27	Uttar Pradesh	9	Barabanki	129	Masauli	949	Lakshbar	B
28	Tamil Nadu	22	Madurai	578	NIH I	NIH I	D I I M S T Andinatti	N

44 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 07:01 PM

DATA MODELING

Star Schema

Purpose:

FACT TABLE

Basic Details of Schools_Sample_Data

DIMENSION TABLES

Dim_State

Dim_District

Dim_Village

Dim_School_Category

Dim_Management_Type

RELATIONSHIPS

One-to-Many Relationships

FILTER DIRECTION

Single Direction

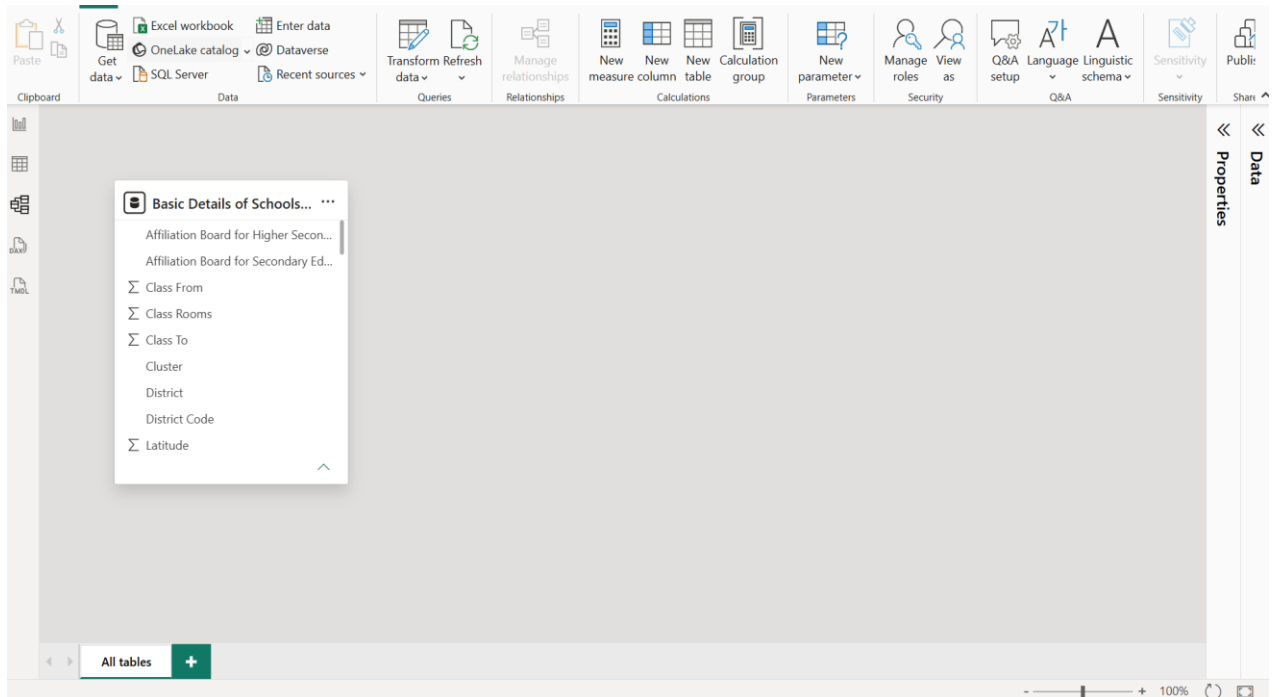
MODEL BENEFITS

Improved Performance

Easy DAX Development

Better Scalability

SCREENSHOT



DAX MEASURES USED

TOTAL SCHOOLS

Total Schools = `DISTINCTCOUNT([State Code])`

PURPOSE : Calculate total schools.

TOTAL STUDENTS

Students = `SUM([Total Students])`

PURPOSE : Calculate total student enrollment.

TOTAL TEACHERS

Teachers = `SUM('Basic Details of Schools_Sample_Data'[Teachers in school])`

PURPOSE : Calculate total teachers.

TOTAL CLASSROOMS

Classrooms = `SUM('Basic Details of Schools_Sample_Data'[Class From])`

PURPOSE : Calculate total classrooms.

AVERAGE STUDENTS PER SCHOOL

Avg Student Per School = $\text{DIVIDE}([\text{Students}], [\text{Total Schools}])$

PURPOSE : Calculate average students per school.

TOTAL STATES

Total States = $\text{DISTINCTCOUNT}(\text{'Basic Details of Schools_Sample_Data' [State]})$

PURPOSE : Calculate total states.

TOTAL DISTRICTS

Total Districts = $\text{DISTINCTCOUNT}(\text{'Basic Details of Schools_Sample_Data' [District]})$

PURPOSE : Calculate total districts.

TOTAL CATEGORIES

Total Categories = $\text{DISTINCTCOUNT}(\text{'Basic Details of Schools_Sample_Data' [School Category]})$

PURPOSE : Calculate total school categories.

AVG TEACHERS

Avg Teachers = $\text{AVERAGE}(\text{'Basic Details of Schools_Sample_Data' [Teachers in school]})$

PURPOSE : Calculate average teachers.

MAX STUDENTS

Max Students = $\text{MAX}(\text{'Basic Details of Schools_Sample_Data' [Total Students]})$

PURPOSE : Identify highest student enrollment.

MIN STUDENTS

Min Students = $\text{MIN}(\text{'Basic Details of Schools_Sample_Data' [Total Students]})$

PURPOSE : Identify lowest student enrollment.

TOTAL VILLAGES

Total Villages = $\text{DISTINCTCOUNT}(\text{'Basic Details of Schools_Sample_Data' [Village]})$

PURPOSE : Calculate total villages.

PASS INDICATOR

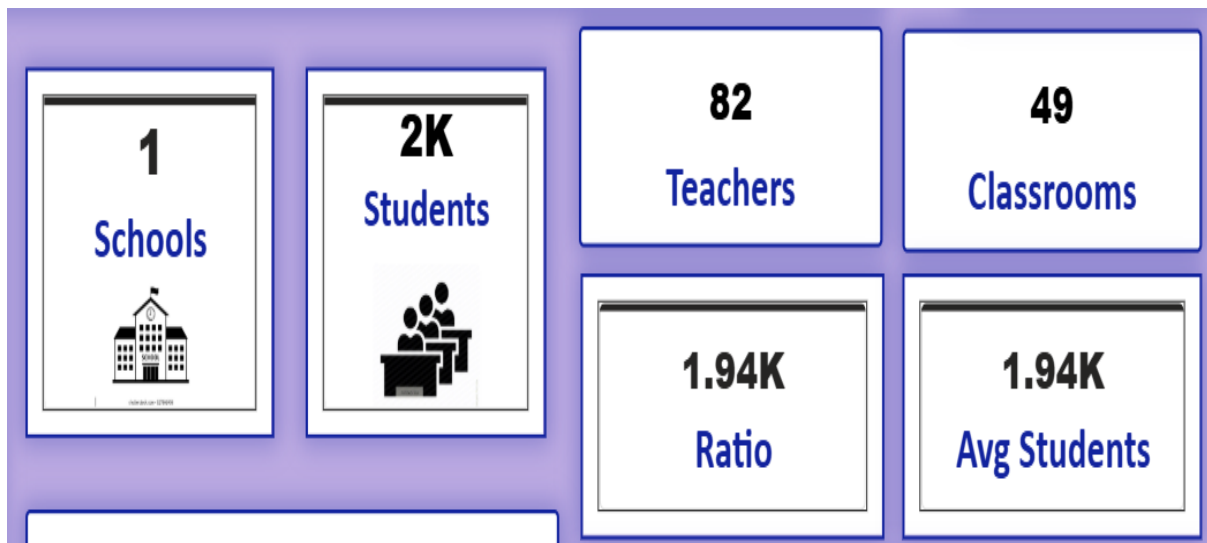
Pass Indicator = COUNT('Basic Details of Schools_Sample_Data'[Status])

PURPOSE : Count school status records.

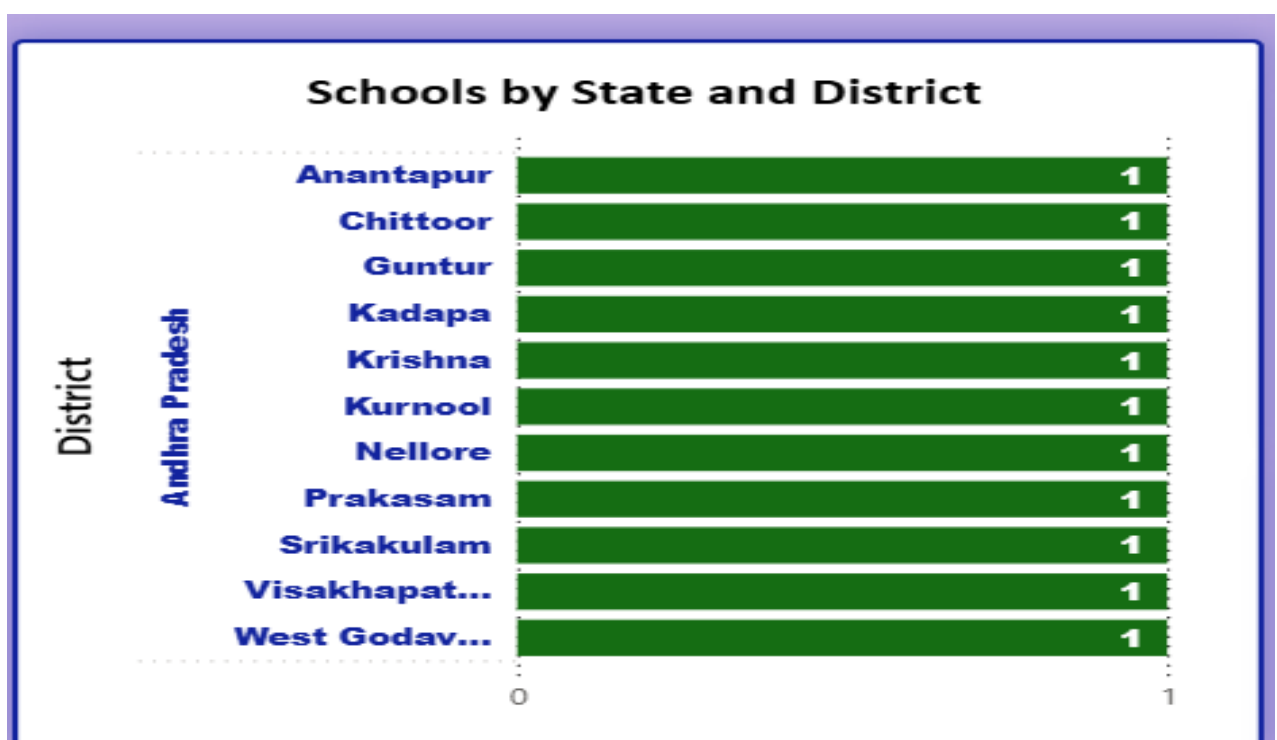
VISUALIZATIONS USED

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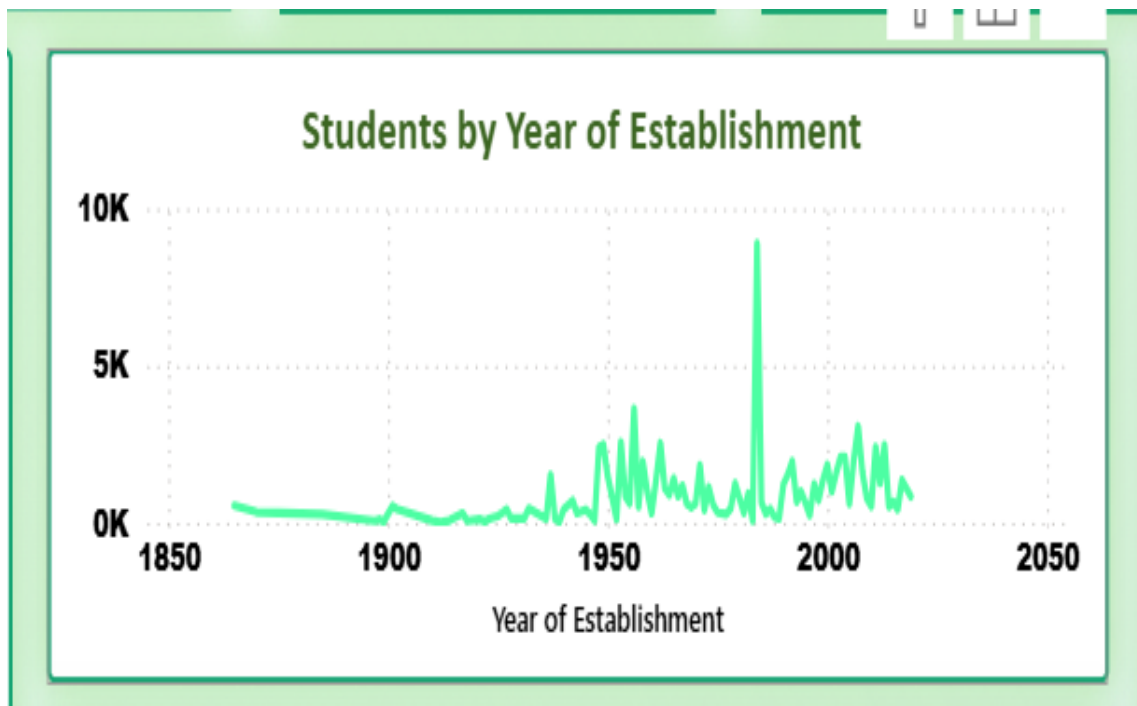
KPI CARDS: Display key educational metrics.



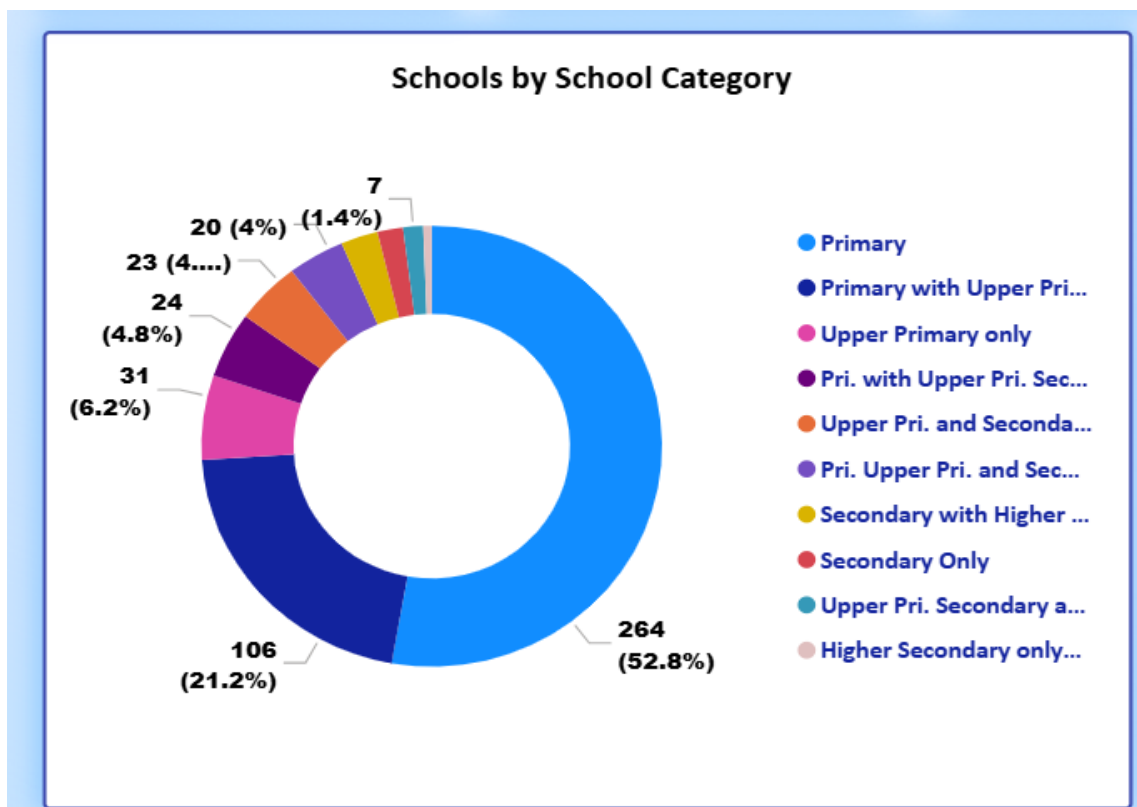
BAR CHART: Compare schools across states and districts.



LINE CHART: Show student enrollment analysis.



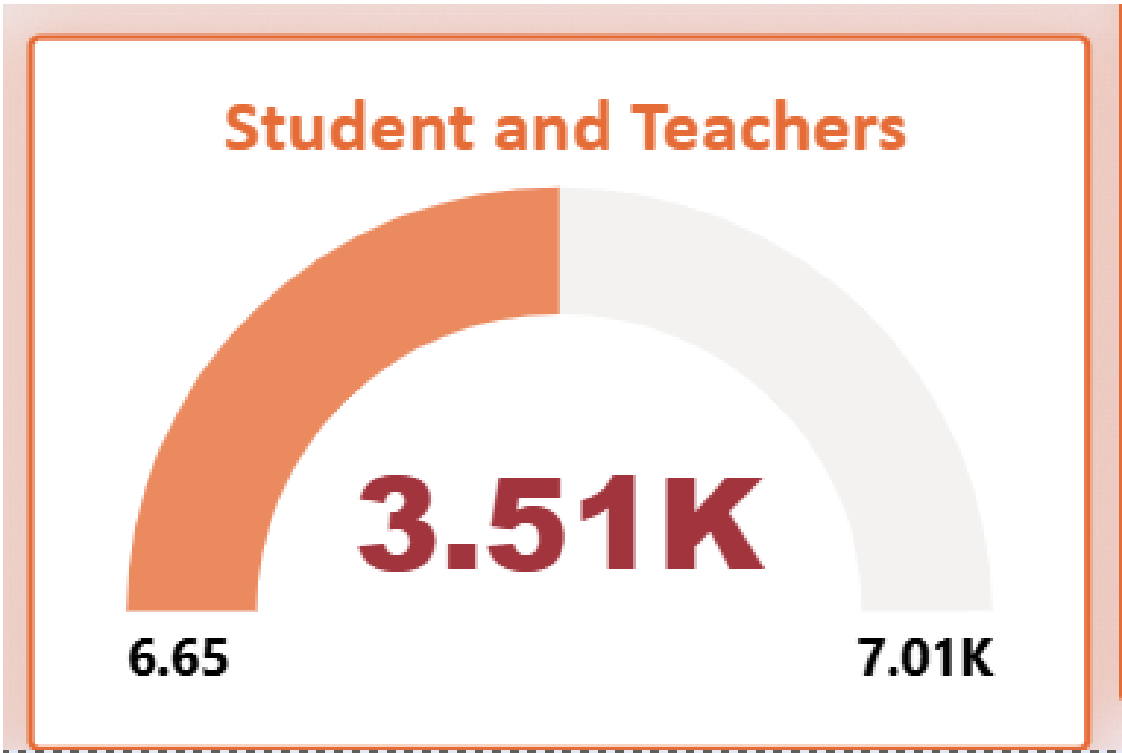
TREEMAP: Show school category contribution.



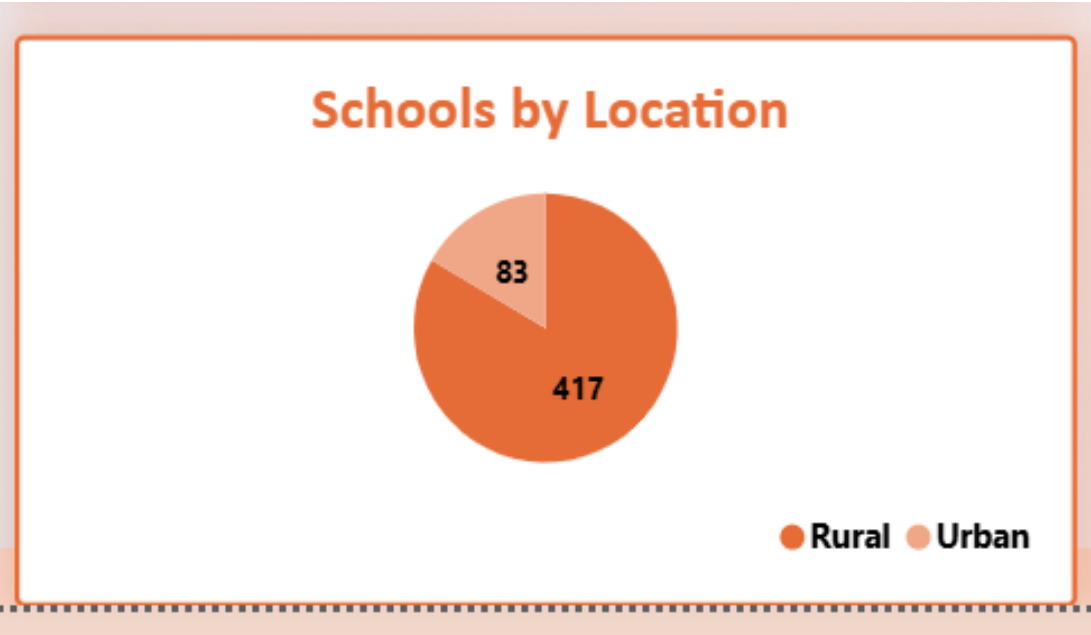
MAP VISUAL: Analyze school distribution by Districts and Villages.



GAUGE CHART: Analyze Teacher-Student Ratio.

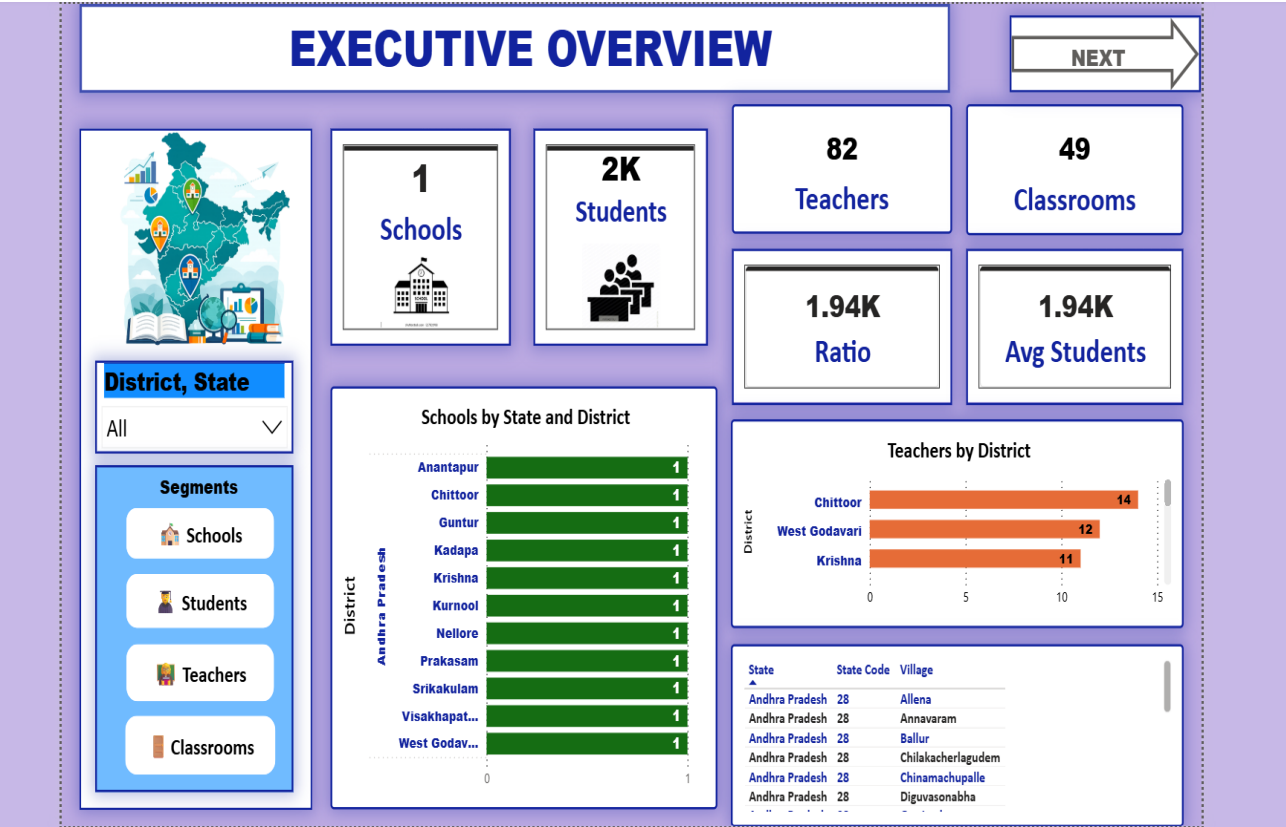


PIE CHART: School Distribution by Location



DASHBOARD SCREENSHOTS

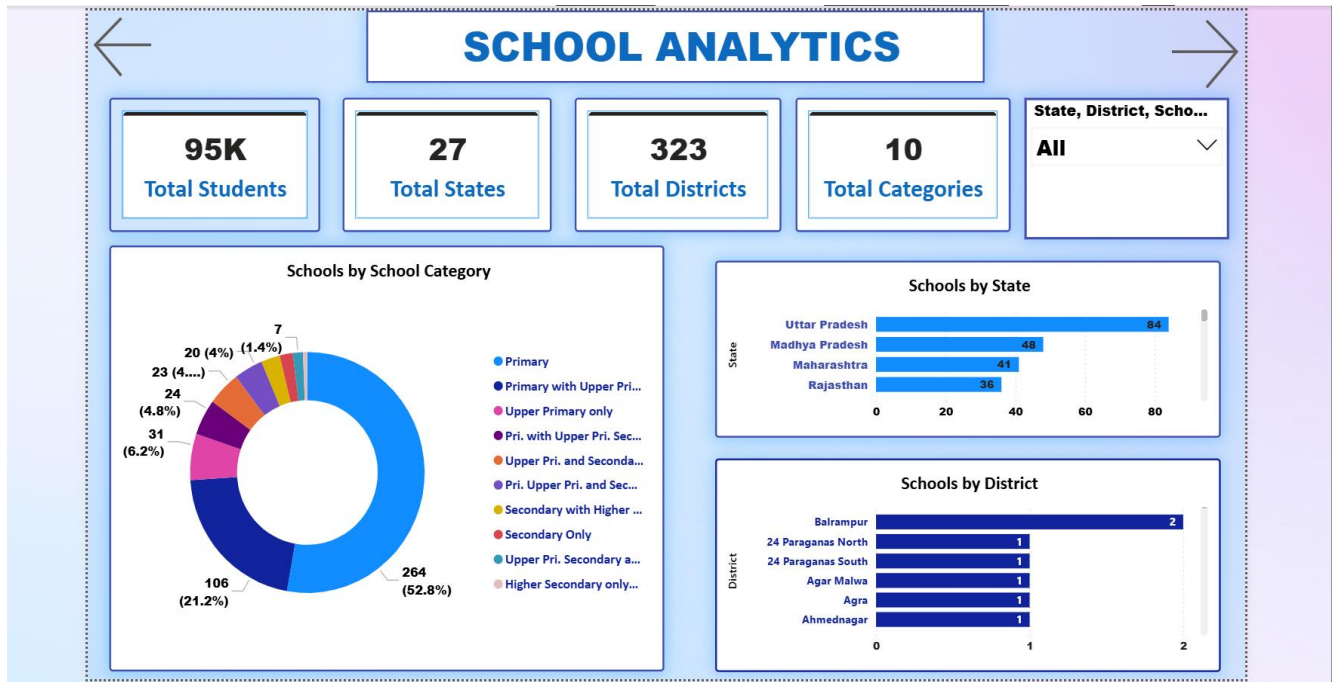
EXECUTIVE OVERVIEW



Explanation

Provides a summary of schools, students, teachers, classrooms, and educational facilities.

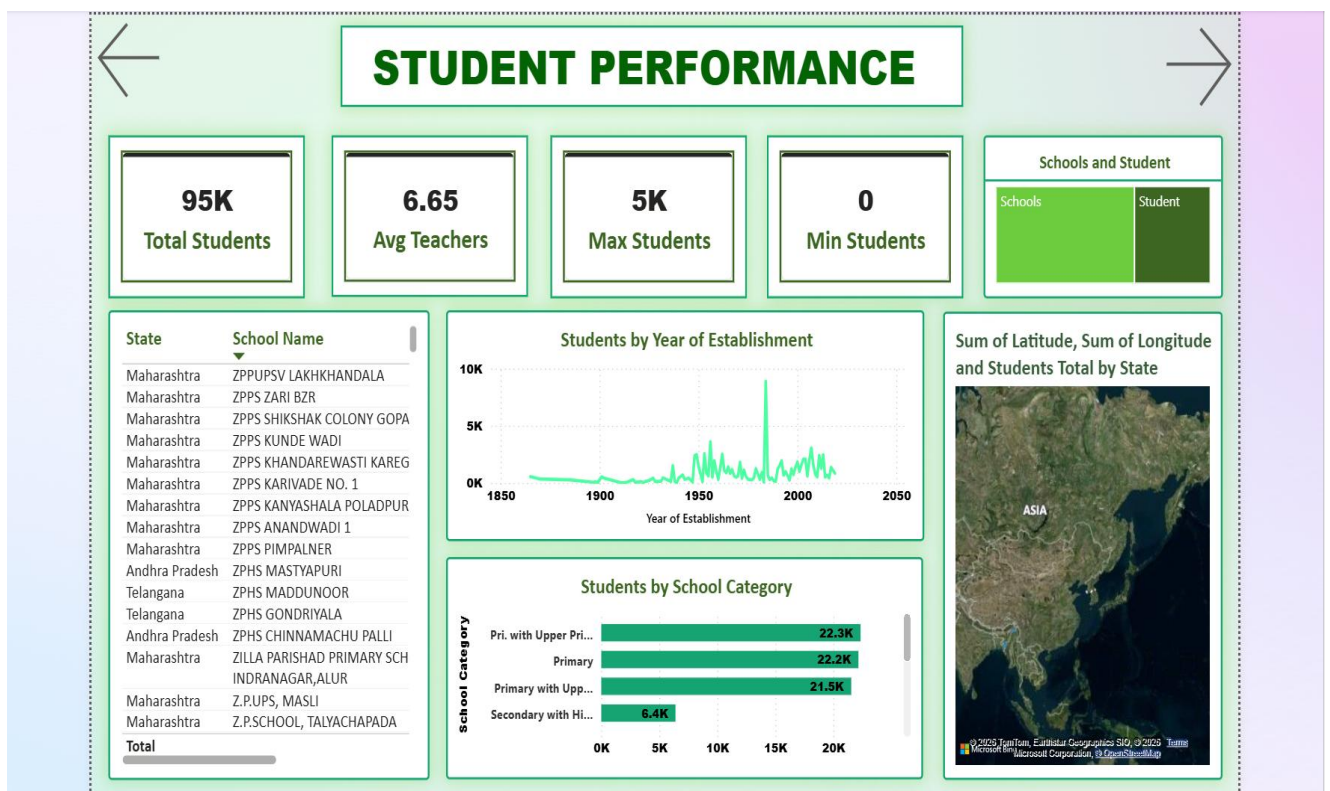
SCHOOL ANALYTICS



Explanation

Analyzes school distribution across states, districts, and categories.

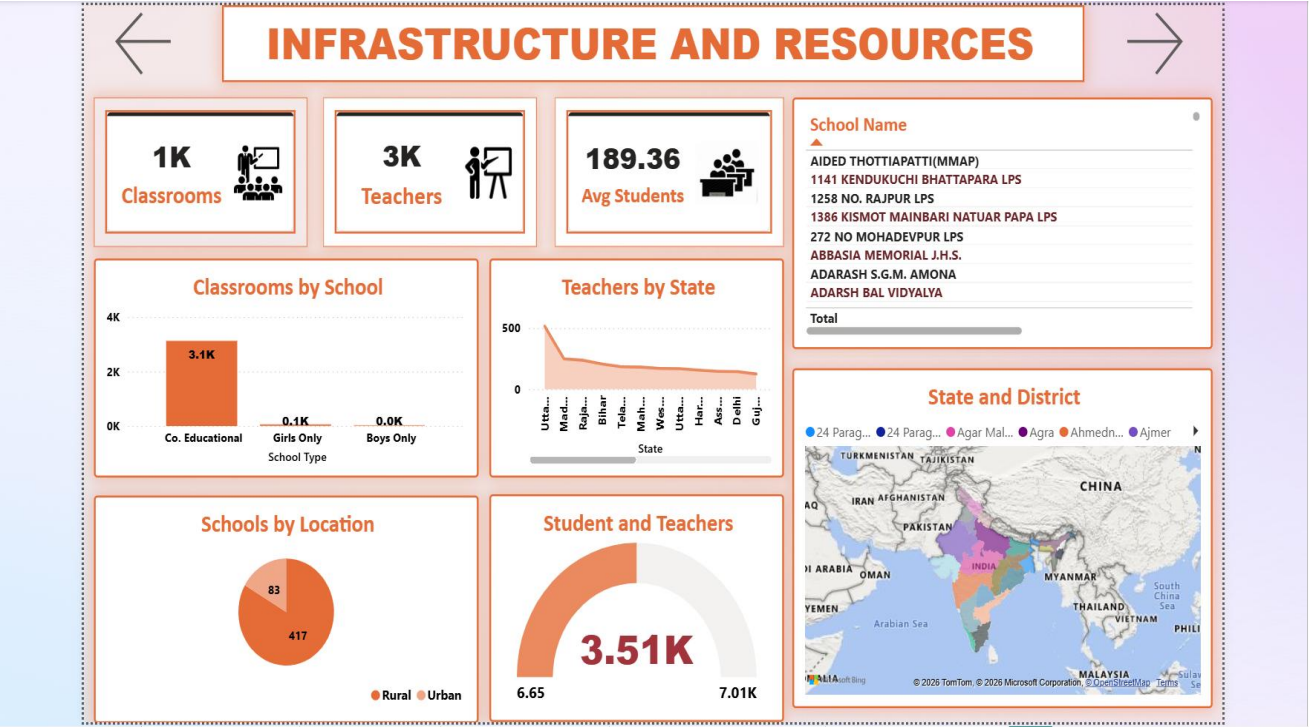
STUDENT PERFORMANCE



Explanation

Analyzes student enrollment patterns and educational reach.

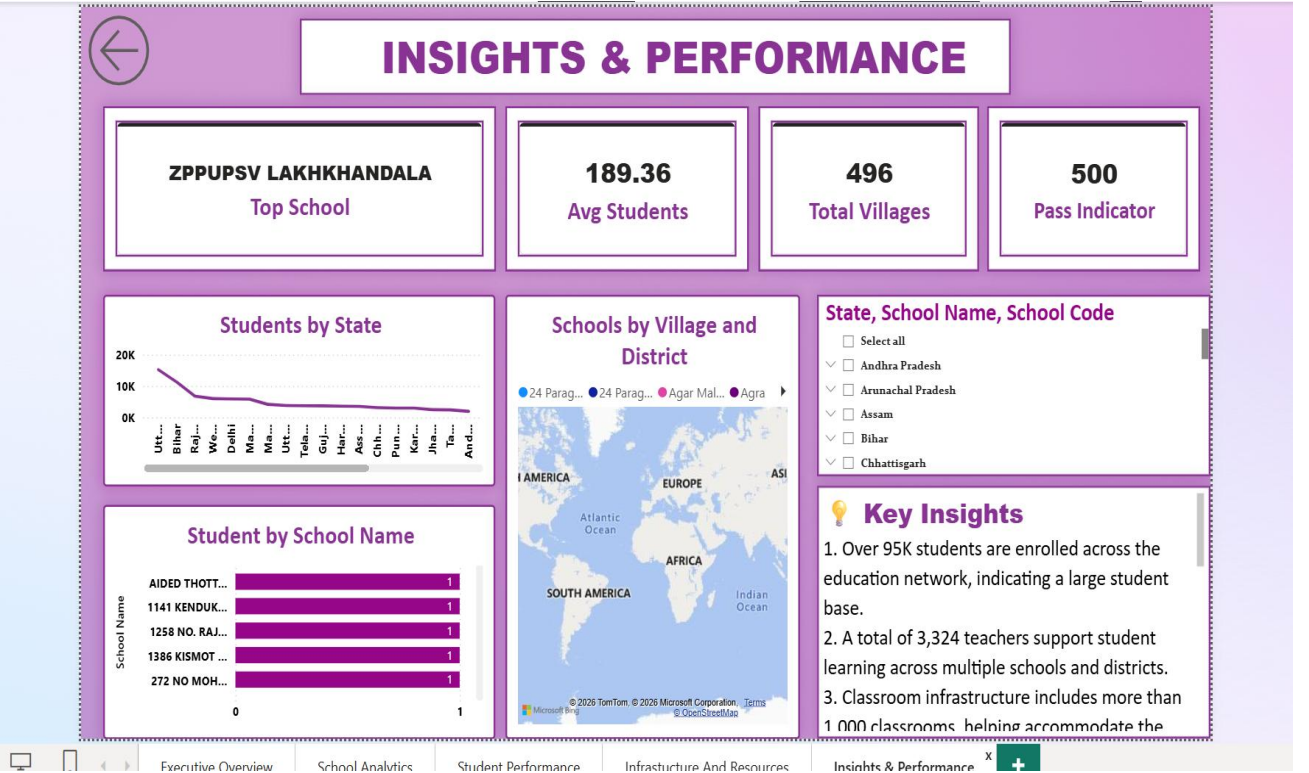
INFRASTRUCTURE & RESOURCES



Explanation

Analyzes classrooms, teachers, facilities, and educational resources available in schools.

INSIGHTS & PERFORMANCE



Explanation

Highlights key performance indicators, village coverage, school status, and important educational insights.

KEY INSIGHTS

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INSIGHT-1

Certain states and districts have a higher concentration of schools compared to others.

INSIGHT-2

Student enrollment varies significantly across school categories and locations.

INSIGHT-3

Teacher availability and classroom infrastructure are unevenly distributed across schools.

BUSINESS RECOMMENDATIONS

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RECOMMENDATION-1

Improve infrastructure facilities in schools with limited classroom availability.

RECOMMENDATION-2

Optimize teacher allocation based on student enrollment and regional requirements.

RECOMMENDATION-3

Strengthen educational resources in districts with high student populations.

RECOMMENDATION-4

Monitor school performance and infrastructure growth for better planning.

FEATURES IMPLEMENTED

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- ✓ Drill Down

- ✓ Drill Through

- ✓ Tooltips

- ✓ Dynamic Titles

- ✓ Slicers

- ✓ Conditional Formatting

- ✓ Navigation Buttons

- ✓ Bookmarks

- ✓ Custom Theme

CHALLENGES FACED

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Data Cleaning Challenges

Relationship Management

DAX Optimization

Dashboard Layout Design

Performance Tuning

LEARNINGS

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Power Query

Data Modeling

DAX Development

Dashboard Design

Storytelling

Business Analysis

PROJECT STORYTELLING

BUSINESS SCENARIO

Educational departments generate large amounts of data related to schools, students, teachers, classrooms, and infrastructure facilities. However, decision-makers often struggle to convert this information into meaningful insights that support educational planning, resource allocation, and infrastructure development.

BUSINESS PROBLEM

- Which states and districts have the highest number of schools?
- Which schools have higher student enrollment?
- How are teachers distributed across different regions?
- Is school infrastructure adequate across all locations?
- Which regions require additional educational resources?

ANALYSIS APPROACH

- Analyzed school distribution across states, districts, and villages.
- Compared student enrollment across school categories.
- Evaluated teacher availability and teacher-student ratios.
- Assessed classroom and infrastructure facilities.
- Identified regions requiring educational improvements.

KEY FINDINGS

States with a higher number of schools generally showed greater student enrollment.

Teacher availability varied significantly across districts.

Infrastructure facilities were not evenly distributed among schools.

Some regions required additional educational resources and classroom facilities.

BUSINESS IMPACT

Supports educational planning.

Improves resource allocation.

Enhances infrastructure development strategies.

Provides data-driven educational insights.

Enables better decision-making for administrators and policymakers.

PROJECT DEPLOYMENT

DEPLOYMENT OBJECTIVE

Deploy the Power BI solution to enable educational stakeholders to access interactive dashboards and insights in real time.

DEPLOYMENT PLATFORM

Power BI Service

DEPLOYMENT PROCESS

Power BI Service

STEP-1

Validate Data Model and DAX Measures in Power BI Desktop.

STEP-2

Optimize Dashboard Layout, Performance, and User Experience.

STEP-3

Publish PBIX File to Power BI Service.

STEP-4

Configure Dataset Refresh Settings.

STEP-5

Create Dashboard from Published Reports.

STEP-6

Apply Security and Access Permissions.

STEP-7

Share Dashboard with Educational Stakeholders.

BUSINESS BENEFITS

Centralized Reporting Platform.

Interactive Education Analytics.

Real-Time Monitoring and Reporting.

Improved Educational Planning.

Faster Data-Driven Decisions.

CONCLUSION

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The School Infrastructure & Enrollment Analytics Dashboard successfully analyzed school distribution, student enrollment, teacher availability, and infrastructure facilities.

The project generated valuable insights that can help educational departments improve planning, resource allocation, infrastructure development, and educational decision-making.